37800: Marketing Management

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SOVALDI™

SOFOSBUVIR

400 mg TABLETS

Investors balked at the deal on Monday, with shares of Gilead falling 9 percent on the announcement.

"For Gilead to give up effectively one-third of their value for an unproven asset still subject to significant ongoing clinical risk seems remarkable," Geoffrey Porges, biotechnology analyst at Sanford C. Bernstein & Company, wrote in a note Monday.

Thomas Wei of Jefferies & Company estimated that Gilead's sales of hepatitis C drugs would have to reach $4 billion a year — difficult, but not impossible — to justify the purchase price.

Healthcare

Gilead to launch generic versions of Epclusa and Harvoni

Sep. 24, 2018 8:42 AM ET | By: Douglas W. House, SA News Editor

- Gilead Sciences (NASDAQ:GILD) announces that newly created subsidiary Asegua Therapeutics LLC will launch authorized generic versions of hepatitis C meds Epclusa (sofosbuvir/velpatasvir) and Harvoni (ledipasvir/sofosbuvir) in the U.S.

- The drugs, priced at $24,000 list to more closely match what the government and health insurers pay, will be available in January 2019. The company says that Medicare Part D patients could save up to $2,500 in out-of-pocket costs per course of therapy.

- Related tickers: (NYSE:ABBV)(NYSE:PFE)(NYSE:GSK)(OTCPK:SGIOF)(NYSE:MRK)(OTCPK:THERF)(NYSE:BMY)(NYSE:JNJ)

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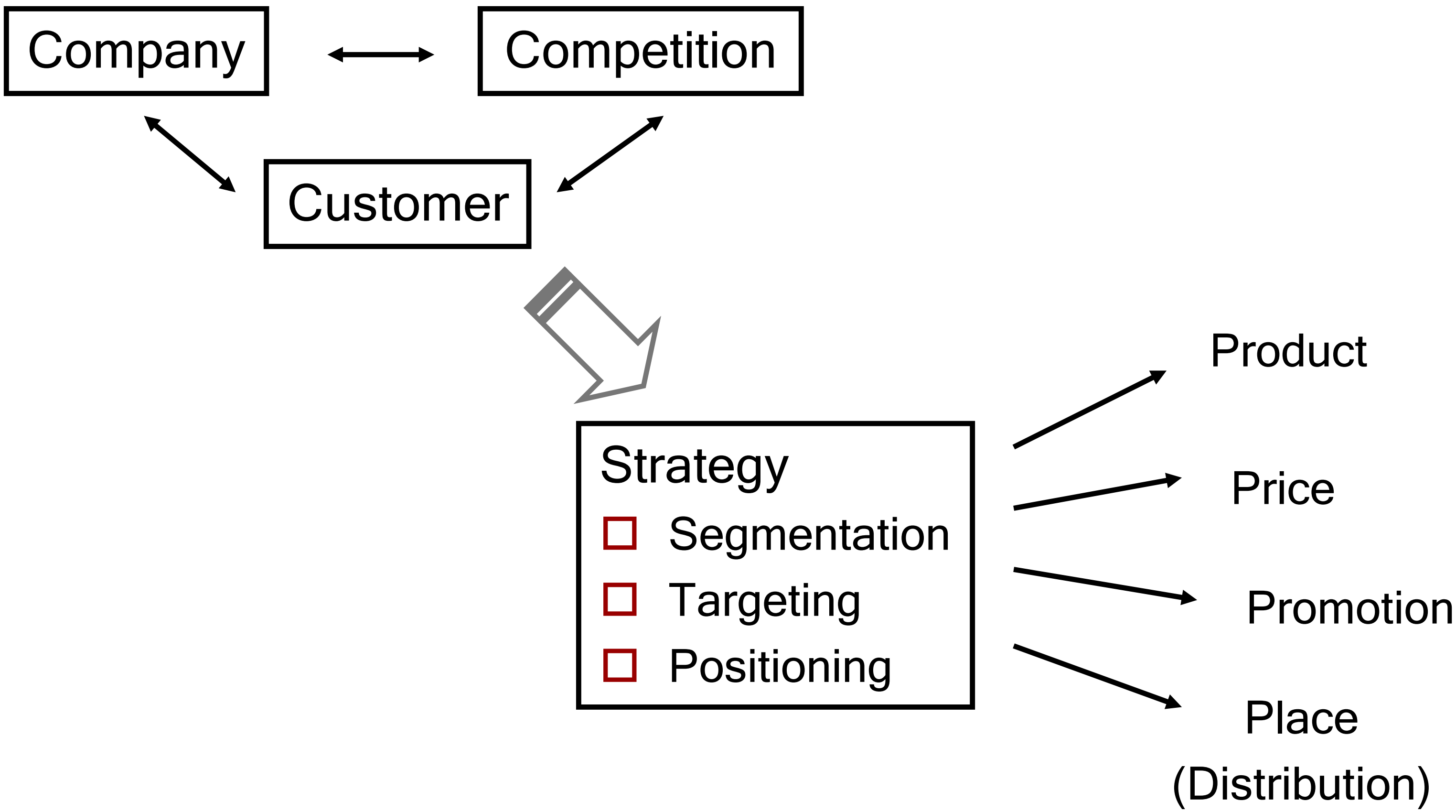
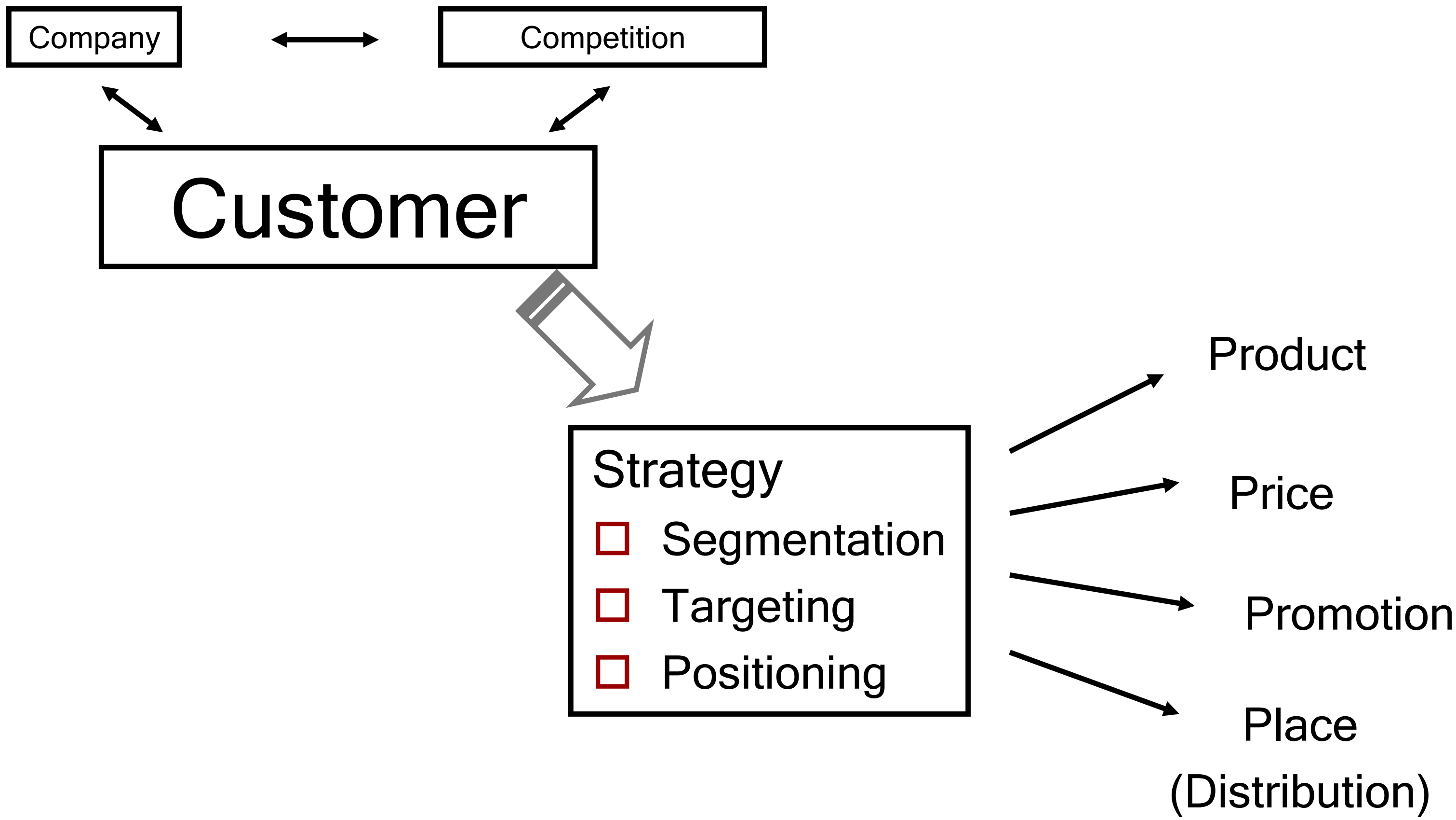
Why did I pick Sovaldi to talk about?

Course Overview

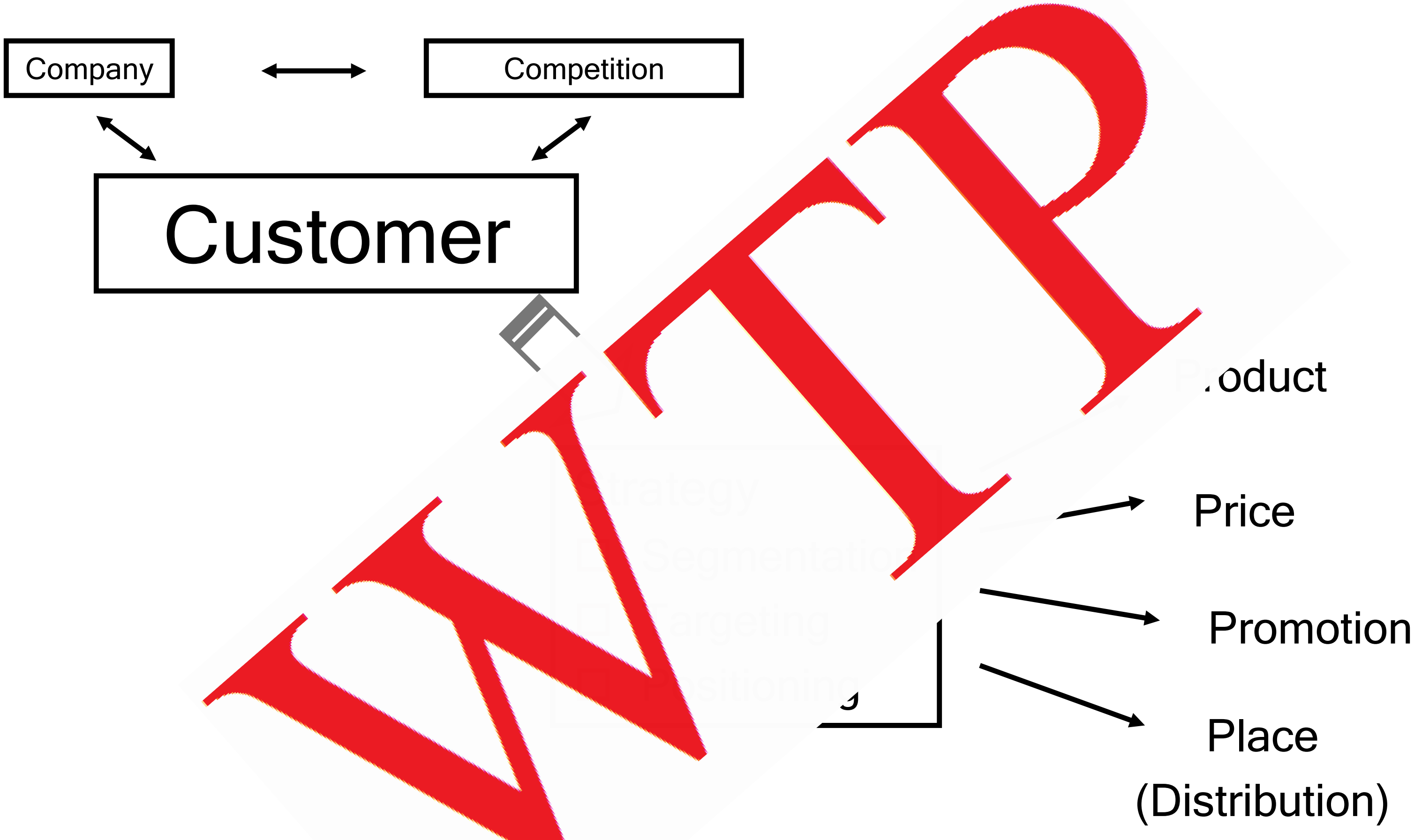
3 C's and the 4 p's



3 C's and the 4 p's



3 C's and the 4 p's



Common "Wisdom" About Marketing

- Commonly thought that marketing is a means of cleverly tricking customers into buying things they don't really want or need.

- This is BAD way to think about marketing.

- Customers have needs → Preferences for product attributes, so those needs can be met.
  - Willingness to Pay over Next Best Alternative

- Our job as marketers:
  - Recognize Needs & Estimate Preferences
  - Design and deliver the product offering

My Goal

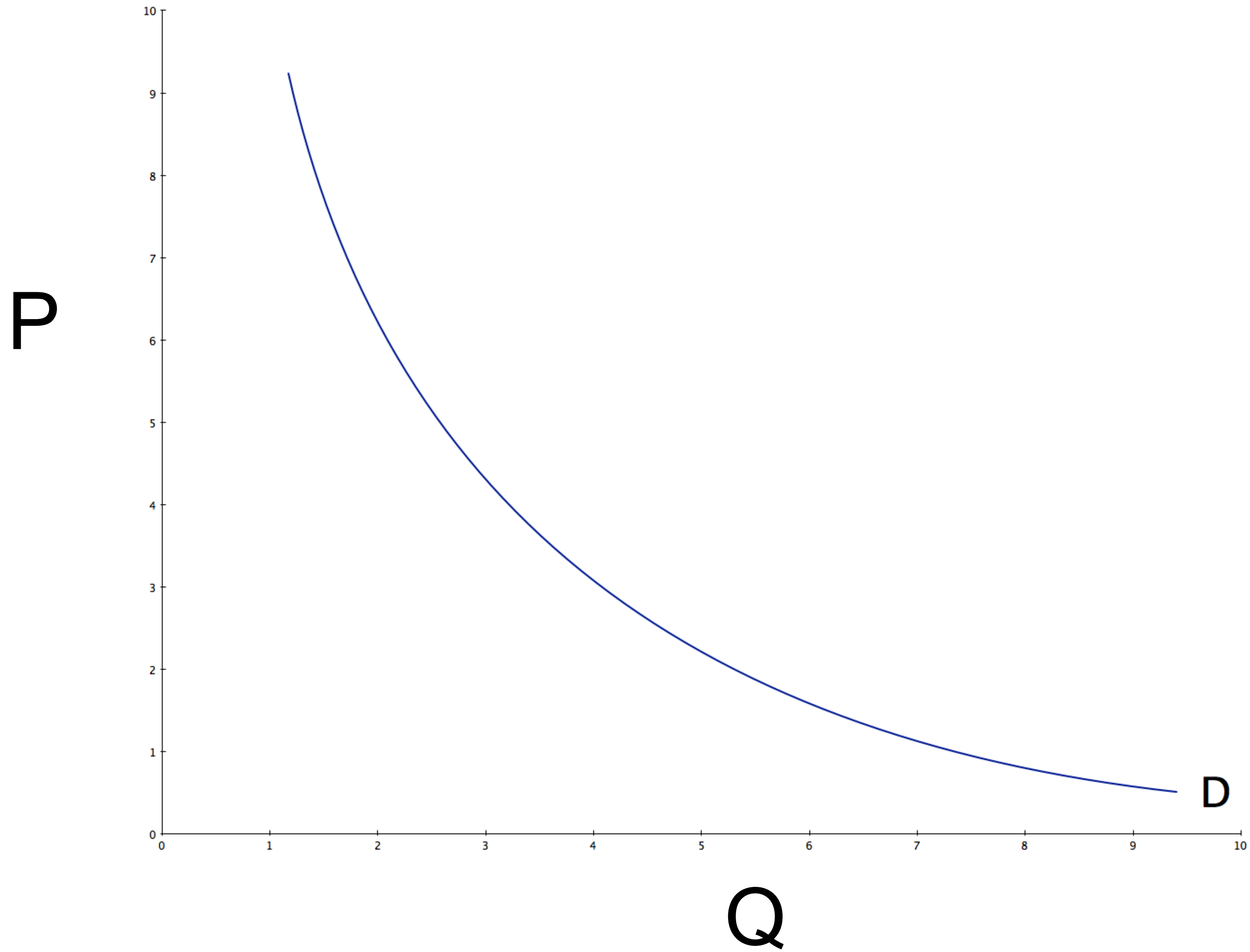
- Take the intuition you learned from Pradeep in Part I
- Apply it in the context of a framework that allows you to get answers.
- He taught you how to think about the three Cs
- Now you need to take that knowledge and *execute*.
  - Price, Promotion, Product Design, Place (distribution, channels, product lines)

Pricing

- We want to set prices to maximize profits.

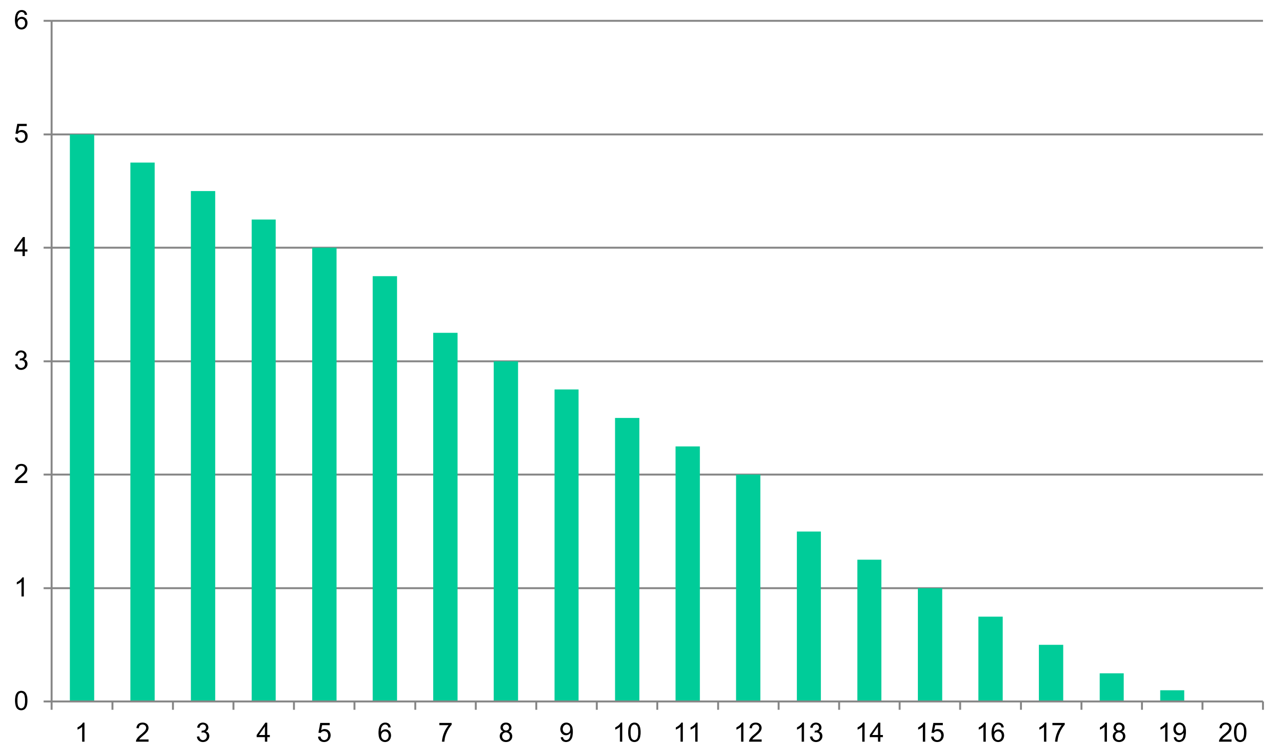
- What is the main tool from micro-economics that helps us do this?

Demand curve



Willingness to Pay

- Imagine that 19 people like good chocolate and their willingness to pay is (highest to lowest):



5
4.75
4.5
4.25
4
3.75
3.25
3
2.75
2.5
2.25
2
1.5
1.25
1
0.75
0.5
0.25
0.1
0

Understanding Demand

- The Good
  – Fundamentally about WTP which is **the** key marketing concept
  – Gives us price elasticities
- The Bad
  – Hard to estimate.
  – Doesn’t tell us anything about individual consumer needs.
  – Doesn’t tell us anything about new products.
  – Doesn’t give us elasticities wrt anything but price

A Practical Framework for Thinking About Consumer Preferences

What is a Product?

What is a Brand?

What is a Product?

- A product is a bundle of attributes
  - Sometimes real and sometimes perceived

- Think about estimating preferences/WTP for *individual attributes* that make up a product
  - Always include price
    - This gives us **better** than price elasticities
  - Brand is included separately
    - This can be especially important with certain product types

Tastes for Attributes

- Consumers have **differing** tastes for these attributes
  - The leads to our segmentation problem.

- Sometimes these are based on **observed** traits and sometimes **unobserved**
- Examples:
  - Size of car:
    - Father of 5
    - Single mid-life crisis
  - Different tastes in beer

Why do we Care?

- Breaking apart the problem
  - No longer about just taste for a product
- Substitution
  - Which consumers?
  - Which competitors?

Competition

- Not just about the characteristics of your product.
  - Also need to think about current and potential competitors
- Sovaldi
  - Which characteristics would you think about?
  - Who are the current and potential competitors?

In Math

$$U_{ij} = \beta_{i1} X_{1j} + \beta_{i2} X_{2j} - \alpha_i p_j + \beta_{i\gamma}\gamma$$

- These betas and alphas are called *part-worths*
  - Intuitively, it is how much this *part* (attribute) is *worth* to customers, measured in utils.
  - Because we have a part-worth on price, we can convert from utils to dollars (we know how much each dollar is worth to customer i in utils)

- Example:
  - Laptops

Example

Weight:	2.9	3.5	4.0
CPU:	50	46	45
Brand:	Apple	Lenovo	Dell
Price:	$1000	$800	$650

Two Types of People

	Type A	Type B
		

Tastes

Weight:	-.3	-.2
CPU:	.4	.02
Brand:	(.5, .1, .1)	(8,3,1)
Price:	-.07	-.02

Negative weights on weight (lighter is better)

Type A is more price sensitive than type B.

Utility of Type A Person

- U = -0.3*(Weight) + 0.4*(CPU) + 0.5 *(Apple) + 0.1*(Lenovo) + 0.1*(Dell) – 0.07*(Price)

- Apple:

-0.3*(2.9) + 0.4*(50) + 0.5*(1) + 0.1*(0) + 0.1*(0)-0.07*(1000)= -50.37

- Lenovo:

-0.3*(3.5) + 0.4*(46) + 0.5*(0) + 0.1*(1) + 0.1*(0)-0.07*(800)= -38.55

- Dell:

-0.3*(4.0) + 0.4*(45) + 0.5*(0) + 0.1*(0) + 0.1*(1)-0.07*(650)= -28.6

Utility of Type B Person

- U = -0.2*(Weight) + 0.02*(CPU) + 8 *(Apple) + 3*(Lenovo) + 2*(Dell) – 0.02*(Price)

- Apple:

-0.2*(2.9) + 0.02*(50) + 8*(1) + 3*(0) + 1*(0)-0.02*(1000)= -11.58

- Lenovo:

-0.2*(3.5) + 0.02*(46) + 8*(0) + 3*(1) + 1*(0)-0.02*(800)= -12.78

- Dell:

-0.2*(4.0) + 0.02*(45) + 8*(0) + 3*(0) + 1*(1)-0.02*(650)= -11.9

Two Types of People

Which is which?

Tying it together

- Basic model assumes people have full information about all products and choose the one with the highest utility
- Other assumptions:
  - Lexicographic
    - Only the most important attribute matters
  - Minimal acceptance
    - First product where all attributes meet a minimum level

Takeaways

- Think about consumer preferences as:
  - Over a bundle of attributes (including brand)
  - Rich consumer heterogeneity
  - Driven by customer needs
- This does two things:
  - Clarifies thinking
    - Helps us solve the real problem – what are the important ones
  - Will help us estimate preferences

Takeaways

- This framework makes marketing strategy an entirely customer-driven (rather than product-driven) exercise.

- You aren't thinking about a product. You are thinking about your customers, what are their needs, and what are the attributes that they are willing to pay for over and above the next best alternative.

You have a utility function, now what?

$$U_{ij} = \beta_{i1} X_{1j} + \beta_{i2} X_{2j} - \alpha_i p_j + \beta_{i\gamma}\gamma$$

- We can use this utility function to:
  - Predict choices (each consumer will pick the product for which they get the highest utility)
  - Determine which attribute is the most important to a consumer or group of consumers.
  - Compute a consumer's Willingness-to-Pay for an attribute

How do I get the part-worths?

$$U_{ij} = \beta_{i1} X_{1j} + \beta_{i2} X_{2j} - \alpha_i p_j + \beta_{i\gamma} \gamma$$

How could we figure out how much a particular customer is willing to pay for one additional unit of fuel efficiency?

How do I get the part-worths?

Chart to show how people change their preferences according to factors / variables.

1. Estimate Using Data

- Option 1: Conjoint! You learned about this in Part I.
  - Just make the product attributes your "bases" variables!
- Option 2: Use field (observational) data
  - We use what we call "revealed preference"

2. Use your specialized knowledge.

- Will show you an example later.

What is Good Variation? Why do I need it?

$$U_{ij} = \beta_{i1} X_{1j} + \beta_{i2} X_{2j} - \alpha_i p_j + \beta_{i\gamma} \gamma$$

- Think about what this utility function means.

- It means *when I increase price*, utility for product j goes down for individual i ***because*** of that price increase.

- This is a **causal** statement.
  - So we need to be able to estimate this function in a way that gives us *causation.*

What is Good Variation? Why do I need it?

$$U_{ij} = \beta_{i1} X_{1j} + \beta_{i2} X_{2j} - \alpha_i p_j + \beta_{i\gamma} \gamma$$

- We're going to measure these part-worths using correlations in data.

- Under what circumstances does correlation imply causation?

What is Good Variation? Why do I need it?

$$U_{ij} = \beta_{i1} X_{1j} + \beta_{i2} X_{2j} - \alpha_i p_j + \beta_{i\gamma} \gamma$$

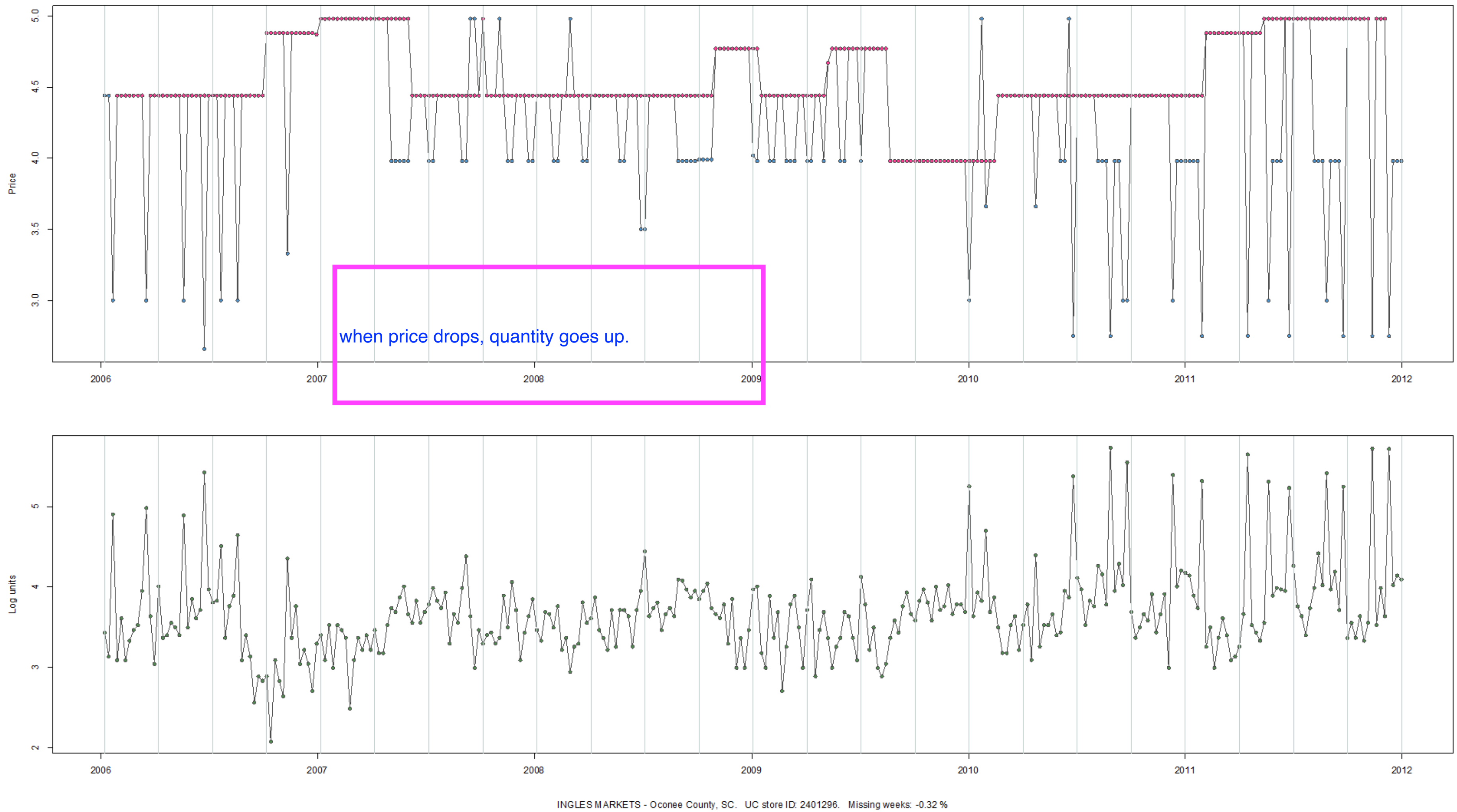
- Does conjoint give us “good variation”? Why or why not?

- Observational data?
  - It depends. It isn’t always provable. We have to convince ourselves of the story.
  - Think about the circumstances under which it *fails*.

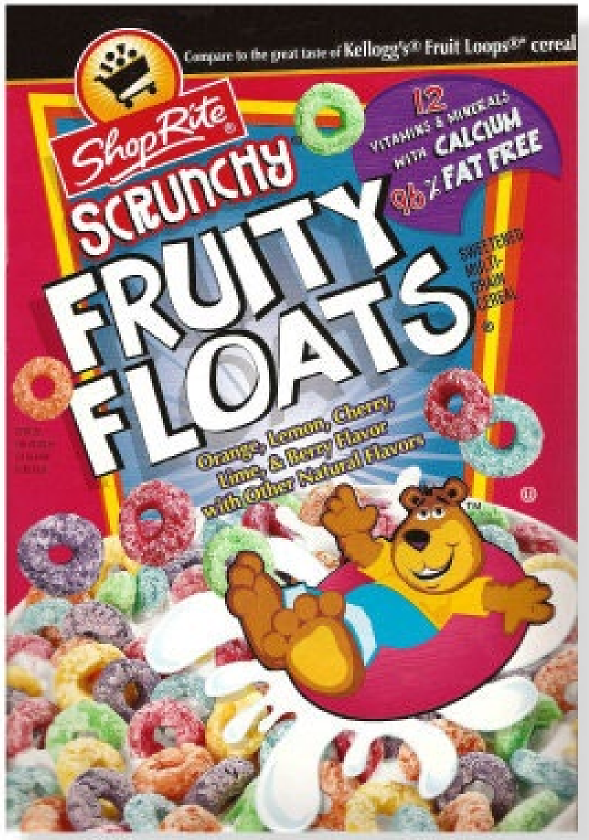
Using Observational Data

- Example of Bad Variation driving estimates:
  - Gasoline during the summer months

- Examples of “good variation”
  - Experiments
  - High frequency promotions (under some circumstances)
  - Will talk more about this concept as we go


Price
when price drops, quantity goes up.
Log units
INGLES MARKETS - Oconee County, SC. UC store ID: 2401296. Missing weeks: -0.32 %

Using Observational Data: Example

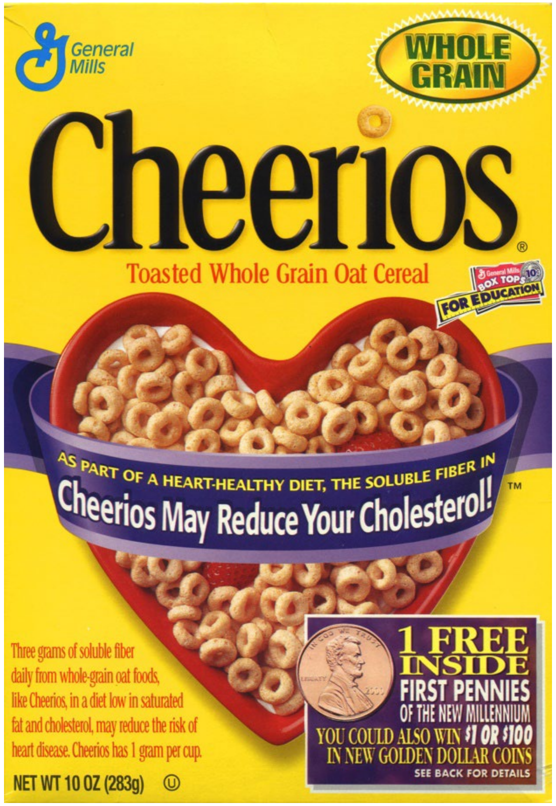


$2.00

Brand

$3.00 $2.75 $2.50

Using Observational Data: Example



Same thing with other characteristics

Using Observational Data

- We then tie these changes in purchase behavior back to consumer observables to incorporate heterogeneity.
  - Demographics / and past purchase behavior
  - Who substitutes to which kind of products as prices change?
  - Who substitutes to new products / characteristics?
- Allows us to map out consumer preferences across different types of consumers
- Since each consumer only makes one purchase, harder to trace out heterogeneity

What if I can’t estimate it using data?

<u>Use your specialized knowledge of customer needs.</u>

- What problem does each attribute solve for the customer?
- How much is that worth to the customer?
- You should always be using this information whether or not you have data.

The Value of Data

25mm PGU-48/B Frangible
Armor Piercing Ammunition

F-35 Lightning

The Value of Data

Data

Decision Making Framework

Data Driven Marketing?

- **No**.

- Data is a tool (an important one), but not an end.

use a customer centric framework

- Use data to refine and inform the parameters of your *decision framework.*

Decision Making Framework

$$U_{ij} = \beta_{i1}X_{1j} + \beta_{i2}X_{2j} - \alpha_i p_j + \beta_{i\gamma}\gamma$$

- Parameters → causal factors impacting demand.
  - Prior beliefs (your specialized knowledge)
  - Data adds precision.
- Combine with info about your cost structure and competitors to write down a profit function.
- Maximize profit.

About Me (and Part II, broadly)

- B.S. Mathematics, B.A. Economics, M.S. Mathematics, Virginia Tech
- PhD Economics, MIT
- Practical
  - Wine importing and sales firm
- Research Interests:
  - Measurement of advertising effectiveness
  - Using marketing tools to answer policy questions
  - Healthcare Industry

What's New

- I am very different from Pradeep.
- We will use a little more math. A little more economics.
  - Intuition is still the most important thing. But we want to arrive at *precise answers* when we can.
- I will send fewer, shorter emails.
  - Value in "struggling through" problems on your own…

Logistics

Structure

- Everything you need to know is in the syllabus.
- Sara is here to help if something isn't clear, but do check the syllabus first.
- The role of the TA is different in Part II than it was in Part I.
  - 2 structured review sessions.
  - Grading. Discussion after.
  - Less evening assistance with assignments due.

Discussion Portion of Class

- Open discussion – mostly today and tomorrow
  - Critically evaluate proposed plans of action
  - Propose alternative viewpoints
- The point is to arrive at the right process for decision making.
  - I will give you my perspective.
- If ever in doubt, always go back to/start from the customer needs.

Lecture Portion

- I will add the structure to the discussion problems so that you have a more general solution of how to proceed.

- You should feel free to contribute ideas from your work experience on the topics.

- The write-up assignment for session 8 will involve discussion that is mixed in with the lecture.

My philosophy about grades

- I am generally unconcerned about your grades.

- I want you to learn the concepts of the class.
  - You’re all spending your valuable time to do this.

- Most of your grade will be “effort” related items.
  - I’m quite serious about this part, though.
  - Show up late, don’t show up, don’t contribute to class, phone in your assignments – these will all hurt you a lot.

- If you put in a lot of effort to the class, you’re not going to get a “bad” grade. Exam will largely separate “A” from “B”.

Break

37800: Marketing Management

Competitive Pricing Strategy

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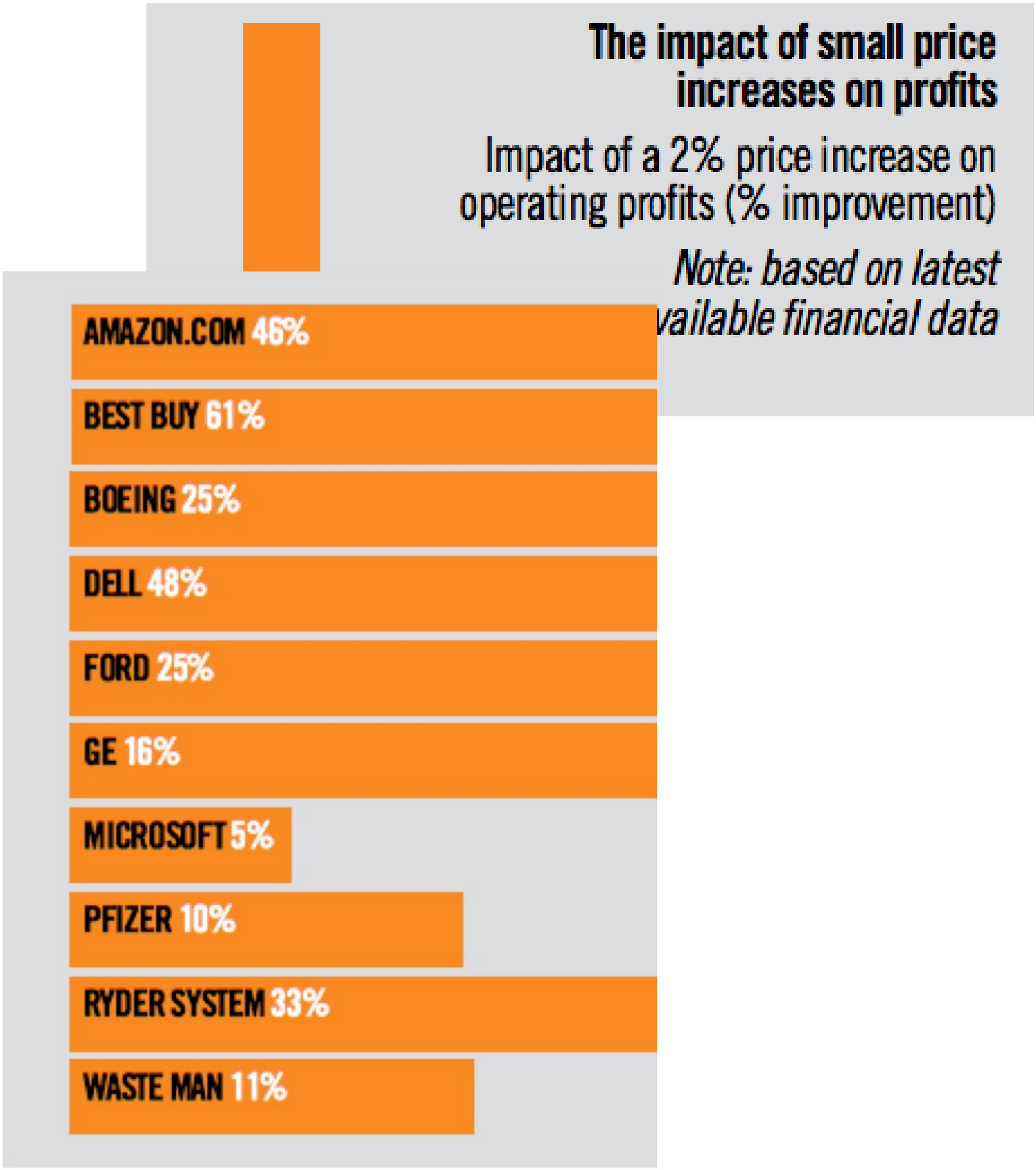
Why I love pricing!

- Makes a huge difference.
- You'll outsource many of your other marketing actions
- Almost every firm sets its own prices!
- Many firms do this very poorly.

Outline for Today

- Optimal Pricing
  - Firm profit function
    - Walk through example
  - Computing Economic Benefits
    - Reference and Differentiation Value
  - Other “rational” factors affecting pricing
- Behavioral Pricing (time permitting)
  - Prospect theory
    - Framing, Loss aversion, Reference pricing
  - Price as a signal of quality
  - Price cues

Pricing has a substantial impact on the bottom line. We looked at ten companies from the S&P 500 and found that a two per cent increase in price increases operating profitability by 14 per cent on average. For companies with low profit margins — such as Amazon.com or Best Buy — the impact of small increases in price on profits can be substantially bigger.


The impact of small price increases on profits
Impact of a 2% price increase on operating profits (% improvement)
Note: based on latest
vailable financial data
AMAZON.COM 46%
BEST BUY 61%
BOEING 25%
DELL 48%
FORD 25%
GE 16%
MICROSOFT 5%
PFIZER 10%
RYDER SYSTEM 33%
WASTE MAN 11%

The Firm Profit Function

- A firm’s goal is to maximize profit by choosing optimal prices
- Two sessions of this:
  – Today assuming we can only set one price per product at one time
  – Tomorrow
    - Multiple prices (price targeting)
    - Prices that change over time (dynamic pricing)

Optimal Pricing

$$\begin{aligned}\text{Profit} &= \text{Revenue} - \text{Cost}\\ &= Q(P)*P - Q(P)*MC - F\\ &= Q(P)*(P - MC) - F\end{aligned}$$

Goal is to choose P to maximize profit

Can think of this in terms of demand function

Optimal Pricing

- But, let's be a little more sophisticated and consider the optimal consumer decision problem
  - Q = # of people who choose our product
- We want to know *which* (segments) consumers and *why* (product characteristics) they choose or don't choose our product.
  - Said differently, Q is a function of p and X
- This approach is based on the "economic value to the consumer" (EVC)

X is product attributes

Optimal Pricing

	WTP	WTP
Consumer 1	3	4
Consumer 2	3	5
Consumer 3	2	6
Consumer 4	2	7
Consumer 5	1	8

$$U = WTP - p$$

Where does that come from?

- Where do their values come from?
  - From the consumer preference estimation
    - Historical data (observational demand analysis)
    - Conjoint analysis

- Who are the consumers we should think about?
  - This comes from the segmentation and targeting exercise in Part I

Optimal Pricing

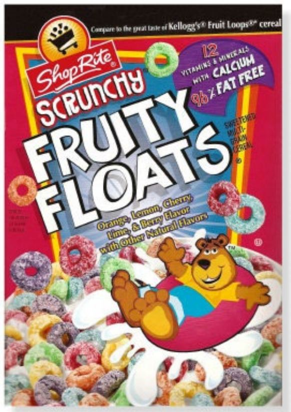
- Who chooses your product depends on:
  - Your prices
  - The prices of your competitors
- Solve the problem for a fixed (current) price that your competitors can charge
- Come back to beliefs about what competitors will do

Optimal Pricing

- Both firms:
  - MC = 1
- Fruity Floats
  - Price = $1
- What price should Fruit Loops set?
  - That is, what is FL's best response to FF's price?

Assume FF is competitive fringe and always charge $1?

Optimal Pricing



Price of Fruit Loops

			1.99	3.99	5.99	7.99
1	3-1	4-p				
2	3-1	5-p				
3	2-1	6-p				
4	2-1	7-p				
5	1-1	8-p				

Optimal Pricing

Price of Fruit Loops

			1.99	3.99	5.99	7.99
1	3-1	4-p	X			
2	3-1	5-p	X			
3	2-1	6-p	X	X		
4	2-1	7-p	X	X	X	
5	1-1	8-p	X	X	X	X
			5	3	2	1

Why not price so that everyone buys?

- Imagine marginal costs are lower than the lowest valuation of your consumers.

- If you charged the lowest value such that everyone bought, you would sacrifice revenue from people with higher valuations
  - Think back to Sovaldi exercise.

- Optimal uniform pricing is always excludes some customers from target segment.
  - Those with relative low WTP.

Optimal Pricing

P	Q	Q*(P-MC)	Profit
1.99	5	5*(1.99-1)	4.95
3.99	3	3*(3.99-1)	8.97
5.99	2	2*(5.99-1)	9.98
7.99	1	1*(7.99-1)	6.99

Optimal Pricing

P	Q	Elasticity	Profit
1.99	5		4.95
3.99	3		8.97
5.99	2		9.98
7.99	1		6.99

Optimal Pricing

E = percentage change in quantity over percentage change in price.

inelastic (less than 1).

P	Q	Elasticity	Profit
1.99	5	~0	4.95
3.99	3	-.398	8.97
5.99	2	-.665	9.98
7.99	1	-1.4975	6.99

Elasticities

- Inelastic demand:
  - Raise price = total revenue goes up
    - Price on all units goes up more than you lose in demand change
- Elastic demand:
  - Raise price = total revenue goes down
- Continue to raise price until you shift from the inelastic to the elastic portion of demand
- General rule: Price on the elastic portion of the demand curve!

Another Approach (Economic Value)

- Recall I brought up your *specialized knowledge.*

- For business (and sometimes) consumer markets:
  – Calculate the material value of each attribute
  – Can use "next best alternative" to think about consumer WTP

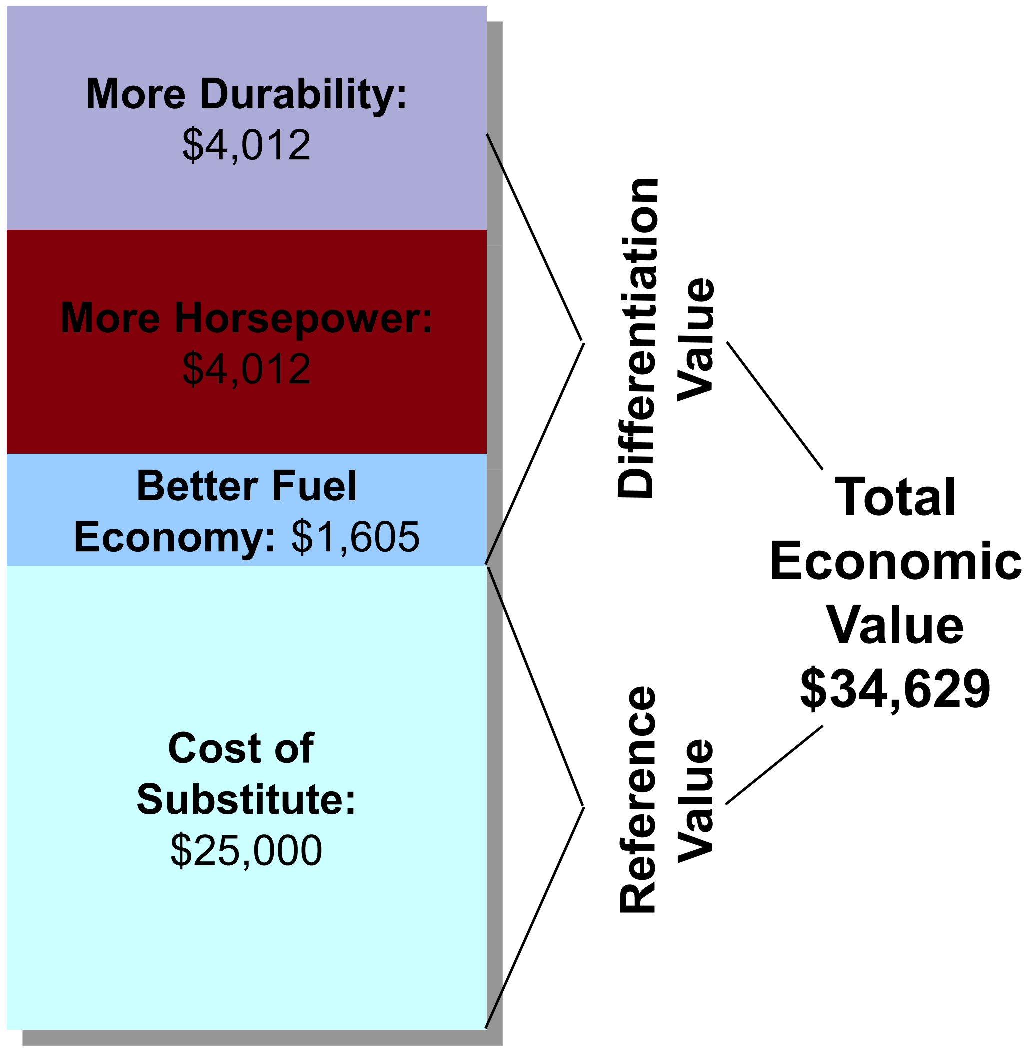
- Fundamentally the same exercise, just with different ways of calculating valuations

Pricing a tractor

John Deere 5075E
Price: ?

Kubota M7040SUD
Price: $25,000

Pricing a tractor



- Cost of substitute = $25,000
- Better fuel economy = $200/yr
- More HP = Faster “tractoring” = $500/yr
- More durable = Lower expected maintenance = $500/yr
- Total life of tractor: 10 yrs
- Sum of values
- Discount rate: .95

Does it matter if Kubota prices its tractors wrongly? no it doesnt matter. this strategy is just to use the next best alternative to set your price.

Differentiation value could be negative

Reference and Differentiation Value

- Benefits of this approach:
  – Easy to explain to purchasers
  – Concrete number
- Downsides:
  – Doesn’t (always) capture intangibles
  – Different valuations for different consumers
  – Multiple “reference” goods might give you different answers

- This is the same fundamental exercise as before
  – You are ultimately trying to assign values to part-worths
  – Then use these to derive WTP/NBA

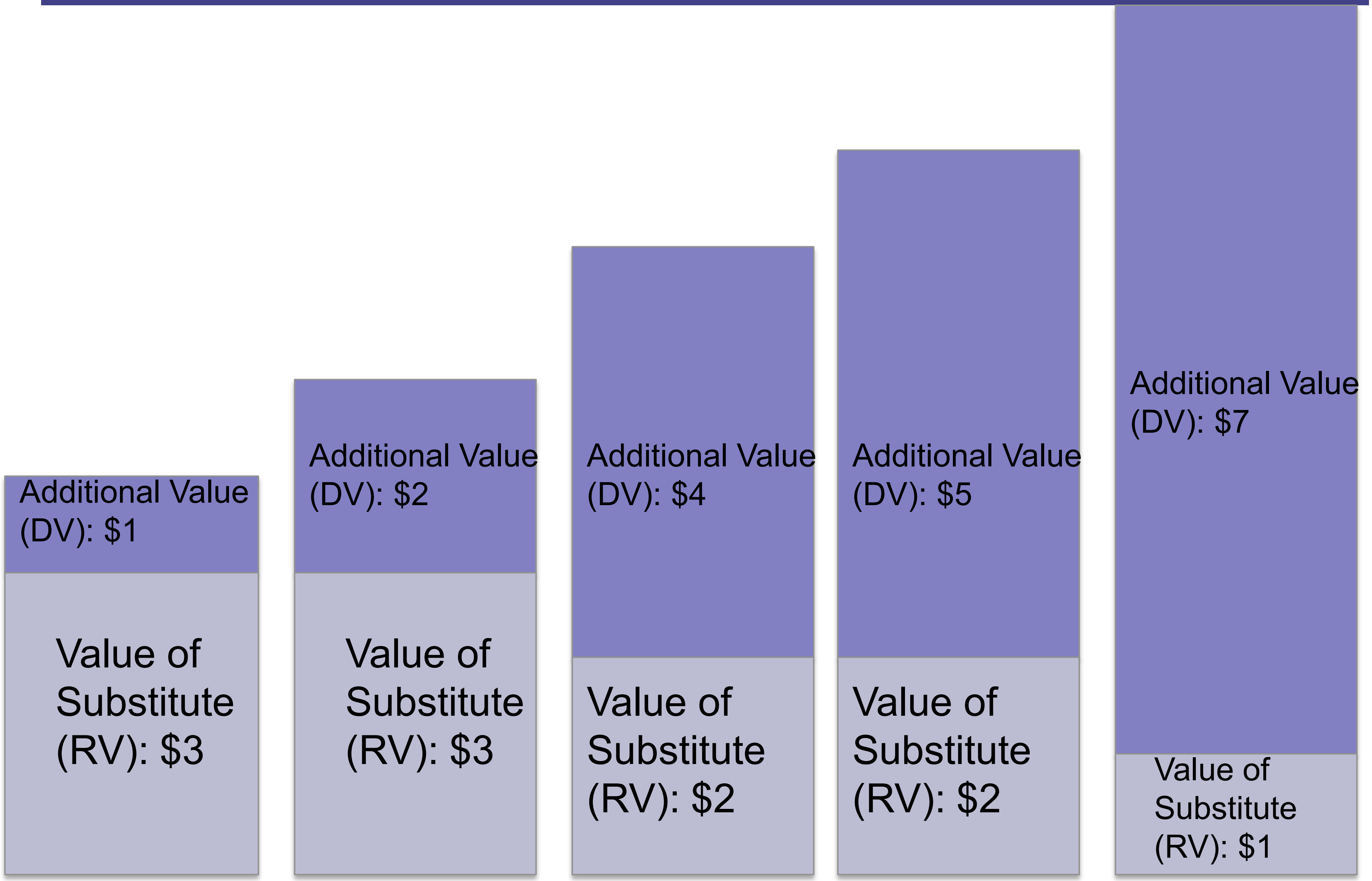
Thought Exercise

- Suppose you can only manufacture one tractor per year.

- What should you be thinking of as your "marginal cost"

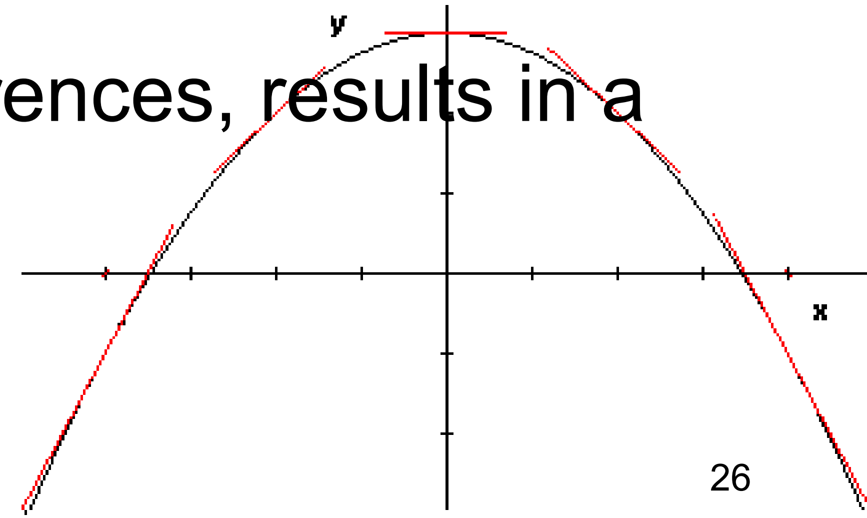
The second highest WTP (?)

Back to Fruity Floats….



This is a Constrained Optimization Problem

- Assign values to part-worths.
- Maximize profit.
  - Subject to the constraint that consumers are choosing their optimal bundle.
- With well-behaved preferences, results in a smooth profit function



More elastic consumer = lower margins

- To maximize current profits, choose price such that profit margin = inverse elasticity.

  Proof (linear cost function):

$$\text{Profit}(p) = p\, q(p) - c\, q(p) - F$$

$$d\text{Profit}/dp = 0 \rightarrow q + p\, dq/dp - c\, dq/dp = 0$$

$$(p-c)\, dq/dp = -q \rightarrow (p-c)/p = -q/dq \,/\, (p/dp)$$

P* - C is the profit.
Fixed cost does not matter to pricing (after differentiating, it is zero).
Elasticity is also a function of price (so E is typically not constand for Q)

$$\rightarrow \quad (p^* - c) / p^* = 1 / |\text{elasticity}|$$

This holds in our framework

- Elasticity is a solved as a function of:
  - Own and competitor characteristics
  - Own and competitor prices

- Can solve the problem holding other variables fixed!

$$\frac{p_j - c_j}{p_j} = \frac{1}{elasticity(\boldsymbol{X}, \boldsymbol{P}, \xi)}$$

How to do this in Excel With Data

- Need data to have good variation.
- Need prices and quantities
- Need some variation in prices
- I'll show you in excel how you would go about doing this.

What is “good variation”?

- Think back to gasoline in the summer.
  - Do people “like” paying more for gas?
  - Why are people driving more when the price is higher?
- Why do we need “good” variation?
  - Utility functions are *causal* statements.

Estimating Price Elasticity: What Ifs

- What if your estimated price elasticity is *positive*?

- What if your estimated price elasticity is *inelastic*?

Going to the data

How Will Competitors React?

- You want to best respond to your beliefs about what your competitors prices will be!
- Another way of thinking about this:
  - What prices should you put into the equation defining which consumers will purchase your product?
- Solve the optimization problem for a fixed competitor price (p)
- Ask the question, what do I believe my competitors will do
  - Starting point is current prices
  - Ending point is iterated best response

General rules for Competition

- The more competitors, the more likely some/all of them will respond
- The competitors with the higher cross-price elasticity (closer substitutes) are the ones that are more likely to respond
- Hold fixed own-price elasticity:
  - 2 competitors with large cross-price elasticities
  - 10 competitors w small cross-price elasticities

Optimal Pricing

- Lesson 1
  - Think about optimal pricing as a constrained optimization problem
- Lesson 2
  - Optimal pricing (almost) always entails excluding some people from the market
    - Trade off extra consumers against higher margin
- Lesson 3
  - Optimal pricing depends on:
    - 1) Consumer preferences (demand)
    - 2) Marginal costs
    - 3) Competition

Outline for Today

- Optimal Pricing
  - Firm profit function
    - Walk through example
  - Computing Economic Benefits
    - Reference and Differentiation Value
  - Other “rational” factors affecting pricing
    - Cost plus and other rules of thumb
    - Fixed vs. variable costs
    - Penetration Pricing
    - Learning by doing
    - Tension with CLV

Example: Park Hannifin

LEADER (U.S.) | March 27, 2007

CHANGING THE FORMULA

Seeking Perfect Prices, CEO Tears Up the Rules

Parker's Washkewicz Weighs Market Power Of 800,000 Parts

By TIMOTHY AEPPEL

CLEVELAND -- In early 2001, shortly after Donald Washkewicz took over as chief executive of Parker Hannifin Corp., he came to an unnerving conclusion. The big industrial-parts maker's pricing scheme was crazy.

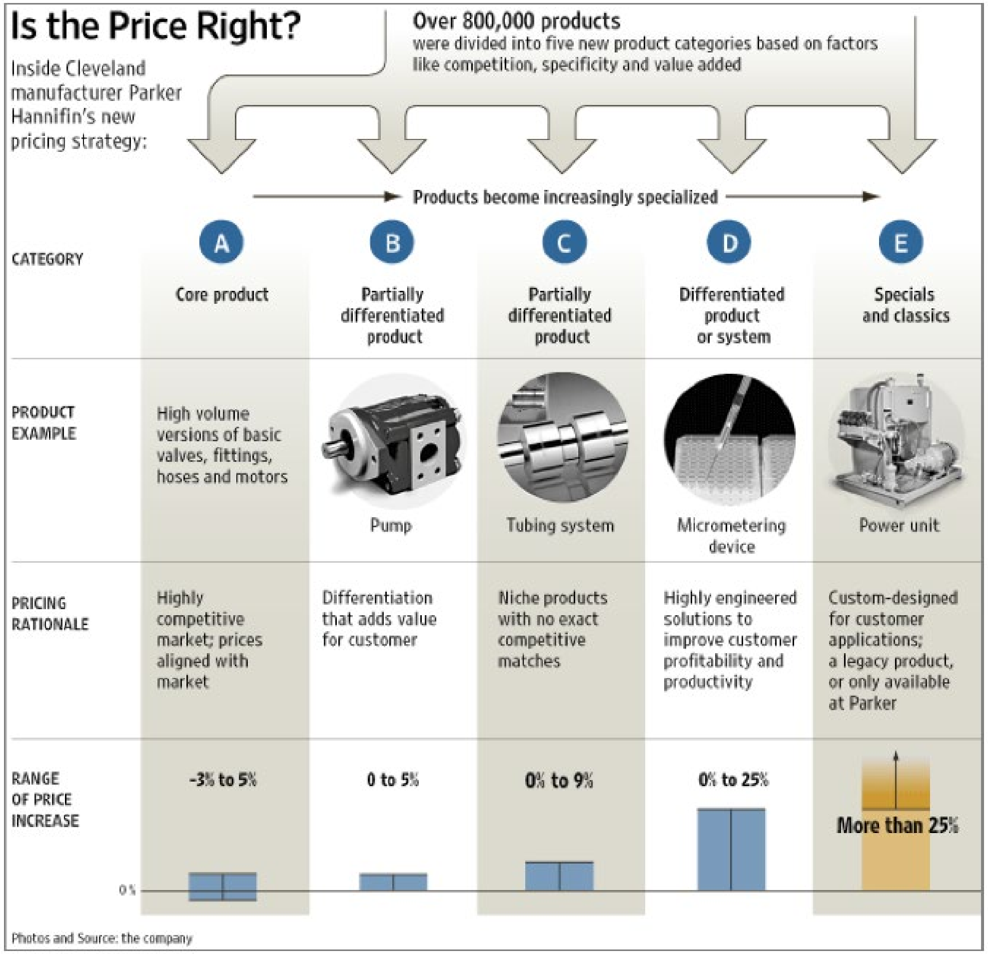
For as long as anyone at the 89-year-old company could recall, Parker used the same simple formula to determine prices of its 800,000 parts -- from heat-resistant seals for jet engines to steel valves that hoist buckets on cherry pickers. Company managers would calculate how much it cost to make and deliver each product and add a flat percentage on top, usually aiming for about 35%. Many managers liked the method because it was straightforward and gave them broad authority to negotiate deals.

But Mr. Washkewicz thought that Parker, which had revenues of $9.4 billion last year, had stuck itself in a profit-margin rut. No matter how much a product improved, the company often ended up charging the same premium it would for a more standard item. And if the company found a way to make a product less expensively, it ultimately cut the product's price as well.

"I was actually losing sleep," recalls Mr. Washkewicz, a 56-year-old Cleveland native who started with the company when he was 22 and rose through its ranks as an engineer.

Cost plus pricing doesnt always makes sense! We should price it base on next best alternative.

Rules of Thumb Are not Necessarily Bad



But they should be based on more than just cost

Cost Plus

- A fixed markup above cost that doesn't depend on competition or demand
  - Uses both fixed and variable costs
- "The third deadly sin is cost-driven pricing. Most American and practically all European companies arrive at their prices by adding up costs and then putting a profit margin on top." Peter Drucker (1996)
- Reasons for it:
  - It's easy
  - If your competitors are doing it
    - Facilitates tacit collusion
  - Transparent / long term fairness
  - It's not that far off
    - Really? Really?
- But, that doesn't mean you should do it!

Another Common Mistake

- Total cost = Q*MC + FC
- Pricing based on fixed costs
  - Since these don't change with quantity, the optimal price is the same with or without them!

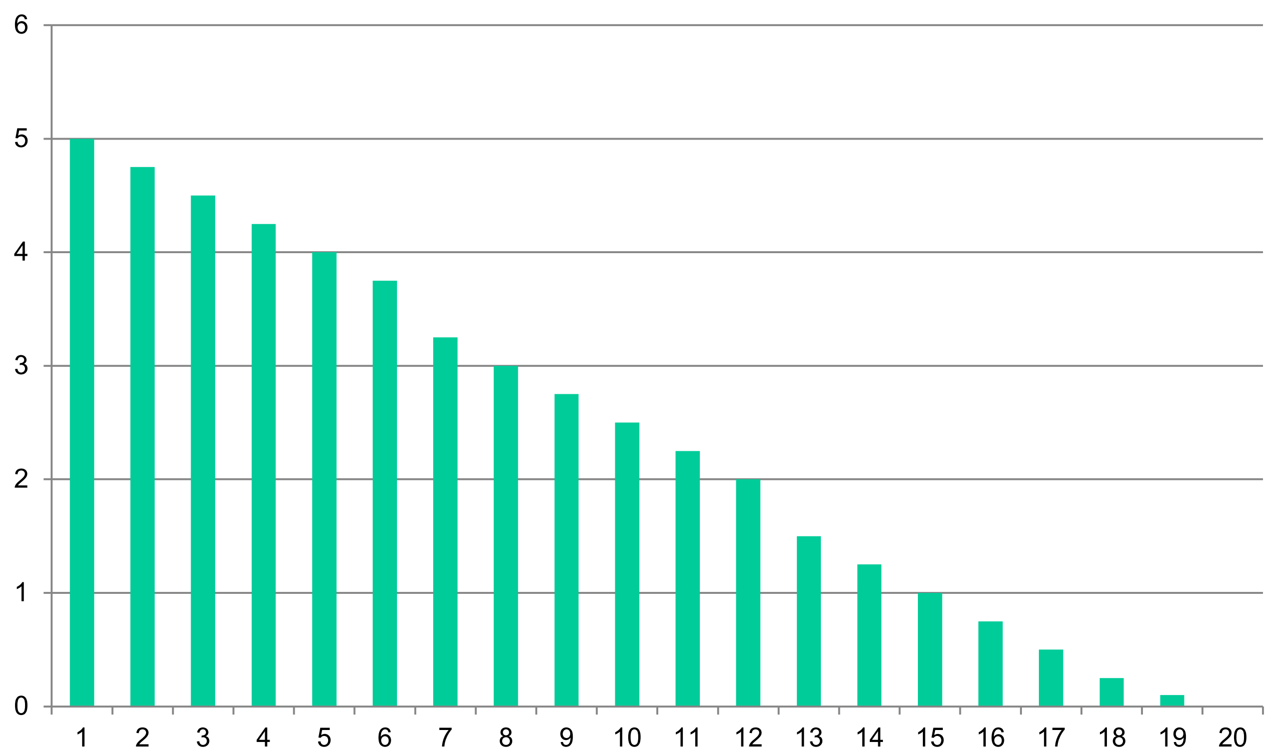
Optimal Pricing – What if there is a fixed cost?

P	Q	Q*(P-MC) - F	Profit
1.99	5	5*(1.99-1)	4.95 - F
3.99	3	3*(3.99-1)	8.97 - F
5.99	2	2*(5.99-1)	9.98 - F
7.99	1	1*(7.99-1)	6.99 - F

Did the optimal price change?

Pricing and Fixed Costs

- F=20, MC=0.5
- Imagine that 19 people like good chocolate and their willingness to pay is (highest to lowest):



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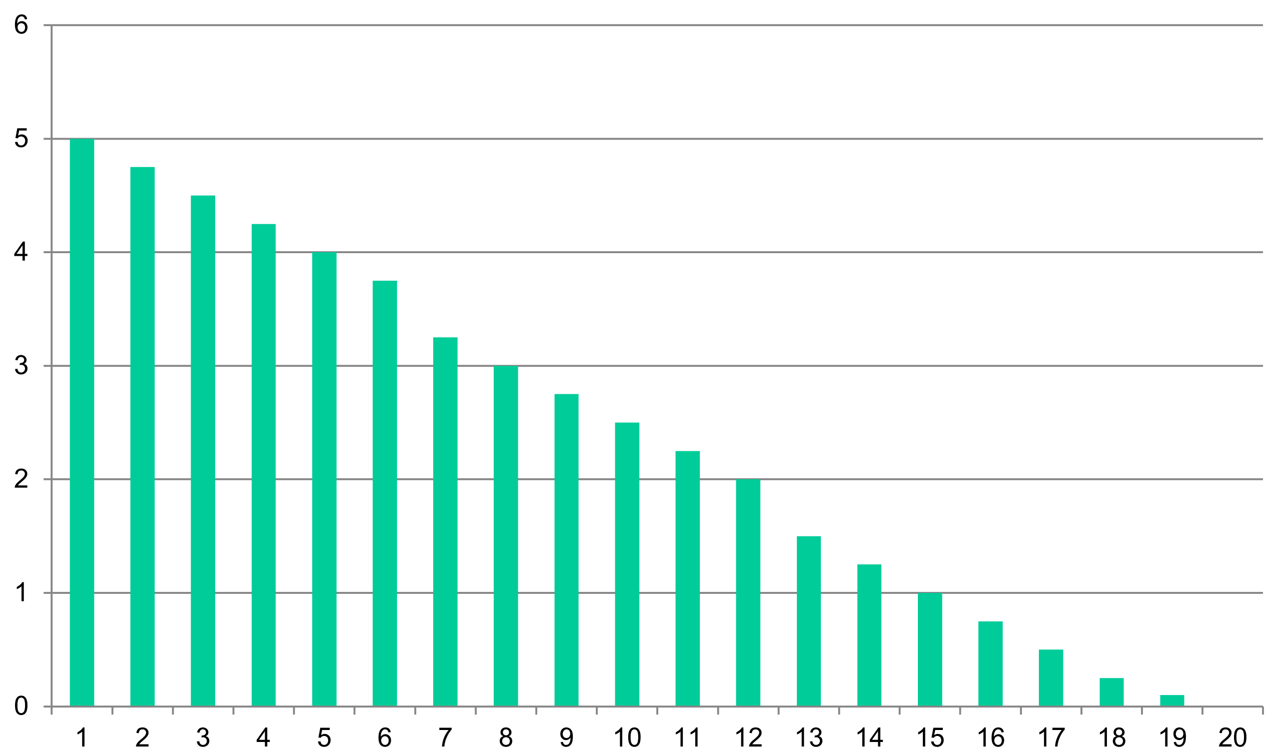
Pricing and Fixed Costs

- F=15, MC=0.5
- Imagine that 19 people like good chocolate and their willingness to pay is (highest to lowest):



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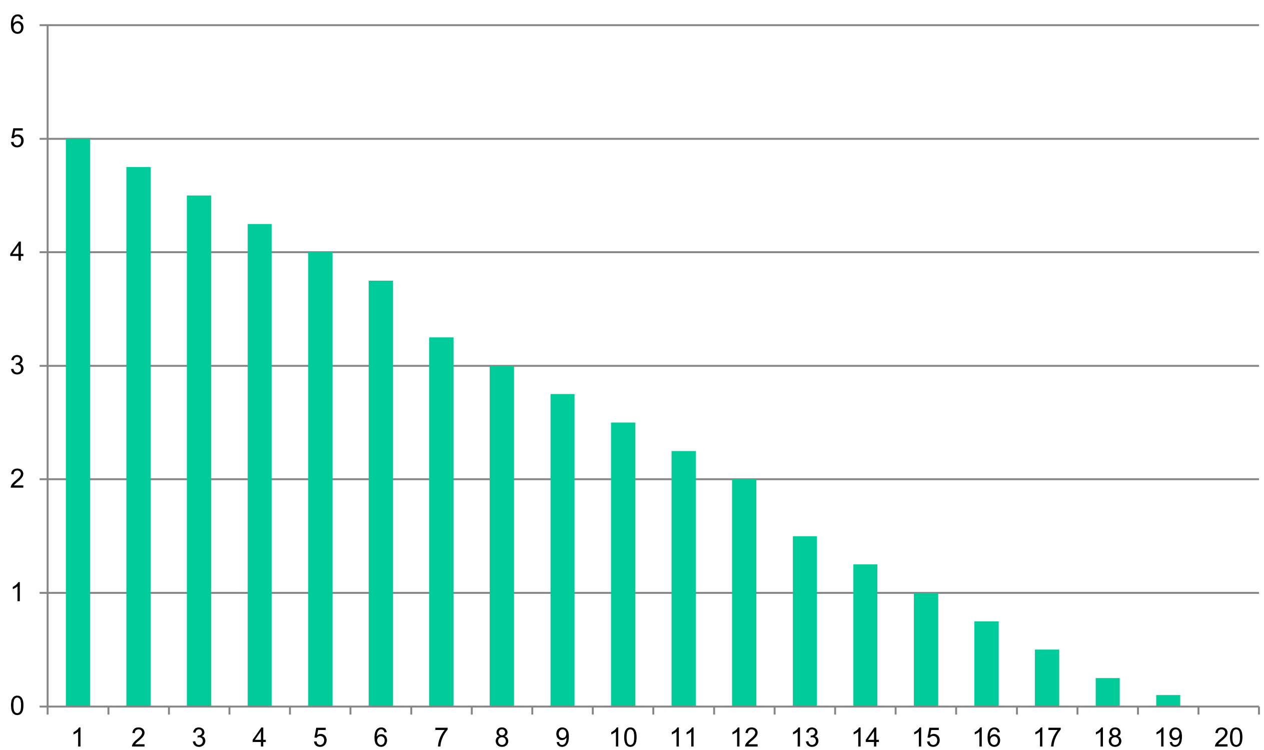
Pricing and Fixed Costs

- F=25, MC=0.5
- Imagine that 19 people like good chocolate and their willingness to pay is (highest to lowest):



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A Common Mistake

- But, how do you make money in an industry with high fixed costs?
  - Fixed costs affect your decision of whether or not to enter an industry or not, but conditional on entering doesn't affect pricing decisions.
- If you can't causally allocate a cost to an individual unit, then it shouldn't matter for pricing

Example: Pharmaceutical Pricing

- Drug costs:
  - Development: $800 million
  - Advertising: $140 million
  - Ingredients: $.05 per pill
  - Factory overhead/maintenance: $10,000 per 100,000 pills
- Only ingredients and (maybe) maintenance matter for pricing.
  - Development, etc, matter for whether to pursue the drug

Other Objectives

- Combination of profit and market share
- “We lose money on every sale, but make it up on volume”
- Pricing lower than optimally in order to “gain market share”
  - Sometimes known as penetration pricing
  - This is only justifiable if there is state dependence (will discuss next week)

“Penetration” Pricing

- Might make sense as part of a longer term strategy
  - But be explicit about this
- Two possibilities:
  - Firm learning by doing
    - Think building airframes
  - Consumer experience / learning
    - Think new CPG products
  - Bottom line: There must be “state dependence”

Consumer Learning Example

Zappos .com

Zappos Zaps Price Protection Policy, Free Overnight Shipping

By consumeristcarey February 17, 2008

Update: Zappos will still surprise customers with free overnight shipping, but it is no longer a perk they will actively promote.

Online shoe store Zappos.com, renowned for their otherworldly 'just-say-yes' customer service philosophy, no longer offers free overnight shipping or a 110% price match guarantee. Tipster Julie forwarded her conversation with a Zappos CSR confirming the policy shift.

Sure enough, there is an official announcement:

> We will no longer be promoting "Free Overnight Shipping" and we no longer will be promoting our price protection policy. Instead, we will be focusing more on our "free shipping" and our expanding selection of merchandise.

75% of customers are repeat buyers

Learning by doing in Production

- Production process where marginal costs get cheaper the more are produced.
- In this case, might want to price low (even below marginal cost) to "walk down the learning curve"
- Creates barriers to entry
  - You can price lower than competitors because your MC are lower

Learning in Production Example

A Dynamic Analysis of the Market for Wide-Bodied Commercial Aircraft

C. Lanier Benkard

+ Author Affiliations

Received May 1, 2000.
Accepted March 1, 2003.

Abstract

This paper uses an empirical dynamic oligopoly model of the commercial aircraft industry to analyse industry pricing, industry performance, and optimal industry policy. A novel feature of the model with respect to the previous literature is that entry, exit, prices, and quantities are endogenously determined in Markov perfect equilibrium (MPE). We find that many unusual aspects of the aircraft data, such as high concentration and persistent pricing below static marginal cost, are explained by this model. We also find that the unconstrained MPE is quite efficient from a social perspective, providing only 10% less welfare on average than a social planner would obtain. Finally, we provide simulation evidence that an anti-trust policy in the form of a concentration restriction would be welfare reducing.

Tension with CLV

The 20 Smartest Things Jeff Bezos Has Ever Said

By Morgan Housel | More Articles
September 9, 2013 | Comments (11)

10. "We've done price elasticity studies, and the answer is always that we should raise prices. We don't do that, because we believe -- and we have to take this as an article of faith -- that by keeping our prices very, very low, we earn trust with customers over time, and that that actually does maximize free cash flow over the long term."

Summary of Optimal Pricing

- Three lessons
  - Think about optimal pricing as a constrained optimization problem
    - Maximize profit subject to the constraint that consumers chose optimally
  - Optimal pricing (almost) always entails excluding some people from the market
  - Optimal pricing depends on consumer preferences, mc, and competition
- Don’t price based on fixed costs!
- Going after market share (for its own sake) is rarely a good idea

Assignments for Tomorrow

- Case memo & discussion: Food Delivery
  - Why are restaurants unhappy?
  - Should we adjust pricing strategy to fix it? If so, how?

- Assignment: Optimal Pricing Using Data (Individual)

Bonus Material – Behavioral Pricing

- Optimal Pricing
  - Firm profit function
    - Walk through example
  - Computing Economic Benefits
    - Reference and Differentiation Value
  - Other “rational” factor affecting pricing
- Behavioral Pricing
  - Prospect theory
    - Framing, Loss aversion, Reference pricing
  - Price as a signal of quality
  - Price cues

How Customers Think

A continuum of views:

Rational Approaches	Bounded Rationality
Standard Economic Theory	Behavioral Economics and Behavioral Decision Research

Bounded Rationality

Customers …

- are only aware of a limited set of products
- have an approximate idea of the value of a product
- are influenced by context and psychological factors
- often use simple heuristics and rules-of-thumb to make decisions

Why does this matter for managers and firms?

- Effects can be sizeable
- Small changes can have significant impact on customer decisions
- Potentially high ROI

Do people act as if they are rational?

"Consider the problem of predicting the shots made by an expert billiard player. It seems not at all unreasonable that excellent predictions would be yielded by the hypothesis that the billiard player made his shots as if he knew the complicated mathematical formulas that would give the optimum directions of travel, could estimate accurately by eye the angles, etc., describing the location of the balls, could make lightning calculations from the formulas, and could then make the balls travel in the direction indicated by the formulas."

Milton Friedman

Departures from “Rationality”

- Do alternate models make sharply different predictions?
- Do these differing predictions matter for marketing actions?
  - In this case for pricing
- Under what circumstances do these alternate models apply?

Outline

- Motivational Example
  - Context matters
- Prospect Theory
  - Reference prices
  - Loss aversion
  - Framing
- Price as a signal of quality
  - Conspicuous consumption
- Price cues

You are lying on the beach on a hot day. All you have to drink is ice water. For the past hour, you have been thinking about how much you would enjoy a nice cold bottle of your favorite beer. A friend gets up to make a phone call and offers to bring back a bottle of your favorite beer from the only nearby place where beer is sold — ***a fancy resort hotel***. He says that the beer might be expensive and asks how much you are willing to spend. He says he will not buy the beer if it costs more than the price you state. What price do you tell your friend?

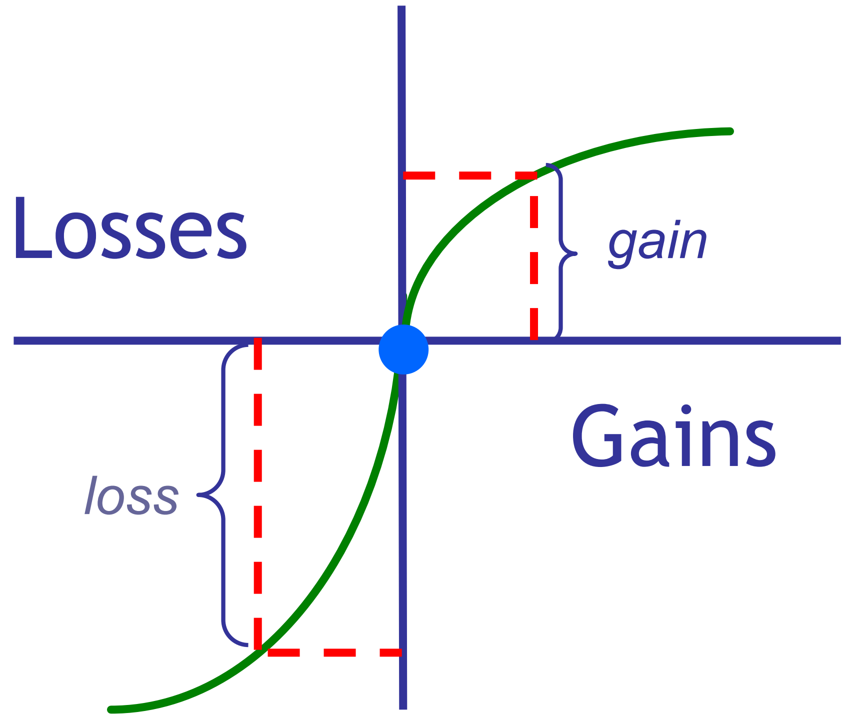
You are lying on the beach on a hot day. All you have to drink is ice water. For the past hour, you have been thinking about how much you would enjoy a nice cold bottle of your favorite beer. A friend gets up to make a phone call and offers to bring back a bottle of your favorite beer from the only nearby place where beer is sold — ***a small, run down-grocery store***. He says that the beer might be expensive and asks how much you are willing to spend. He says he will not buy the beer if it costs more than the price you state. What price do you tell your friend?

Framing / Context

- When Thaler presented these scenarios to a group of executives in the early 1980s, the median response in the grocery store scenario was $6.75 and the median response in the fancy resort hotel scenario was $8.
- Why?
  - in both scenarios, the ultimate consumption is identical — the same beer is consumed on the same beach.
  - no atmosphere from the fancy resort hotel or the run-down grocery store is being consumed by the beer drinker to justify different prices.
  - there is no strategic reason to report a price below one's "perceived value" for the beer [e.g., you cannot haggle over price with hotel or store owner].

Behavioral Approach: Prospect Theory

- Value is judged relative to a reference point
- Losses loom larger than gains
  - Estimates range from 2.0x-2.5x
- Diminishing sensitivity in both directions



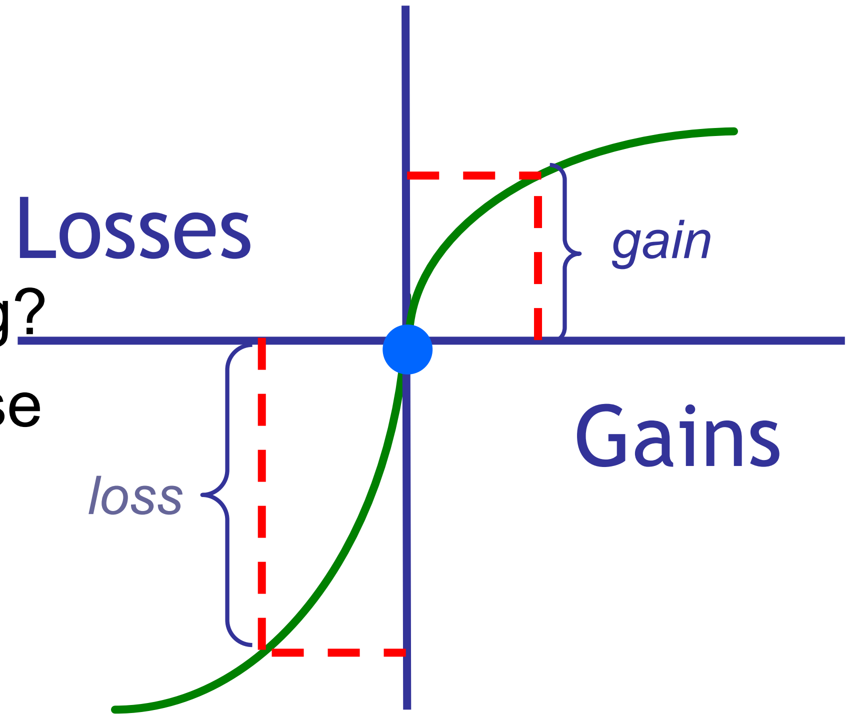
loss > *gain*

Source: D. Kahneman and A. Tversky (1979), "Prospect Theory: An Analysis of Decision under Risk," *Econometrica, 47(2), 1124-31.*

Behavioral Approach: Prospect Theory

- Hedonic Framing
- References Prices
- Loss Aversion
  - How do these affect pricing?

    (And how can you make use of them as a manager?)



loss > *gain*

Source: D. Kahneman and A. Tversky (1979), "Prospect Theory: An Analysis of Decision under Risk," *Econometrica, 47(2), 1124-31.*

Hedonic Framing

- Consider the following scenarios.
  - Jerry is given two lottery tickets. He wins $300 in one lottery and $100 in the other lottery.
  - Elaine is given one lottery ticket. She wins $400 dollars.
- Who is happier, Jerry or Elaine?

Loss Aversion

- Example:
  - Registration for a conference
- The experimental framing manipulation concerned the fee for late registration. Half of the participants were told that there is a “penalty” of $50 for registering after the first deadline. The other half of participants were told that there is a “discount” of $50 for registering before the first deadline.
- When framed as a “discount” only 67% registered early. When framed as a “penalty” 93% registered early.
- Relationships to public policy?

Reference Prices

- External
  - Relative to other (similar) products
- Internal
  - Stored in consumer’s mind. This is usually historical data about the same product.

Reference Prices

- Your favorite sports team has made the playoffs. Its first-round playoff series is a best-of- seven series with Games 1, 2, 5, and 7 played on your team’s home field. General admission tickets had been priced at $20 during the regular season. The team decided to raise general admission prices to $40 for these four playoff games. Is this price increase fair or unfair?
- Your favorite sports team has made the playoffs. Its first-round playoff series is a best-of- seven series with Games 1, 2, 5, and 7 played on your team’s home field. General admission tickets had been priced at $20 during the regular season. General admission tickets were also priced at $20 for Games 1 and 2 of the playoffs. **After Game 2, the team decided to raise prices to $40 for Games 5 and 7. Is this price increase fair or unfair?**

Reference Prices

- In a best-of-seven playoff series between the Miami Heat and New York Knicks, Miami raised ticket prices in the middle of the playoff series.
- For Games 1 and 2, Miami Heat management had set the prices for various seats at $20, $30 and $40. After Game 2, the Heat raised ticket prices to $50, $80 and $90 for Game 5.
- Public outrage resulted and Game 5 was one of the very few playoff games that year not to sell out.
- The extent of the outrage forced Miami Heat management to return prices to $20, $30 and $40 for the final and deciding Game 7.

Reference Prices

Hot New Releases › See Top 100

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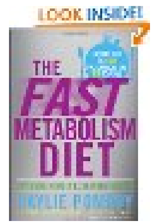
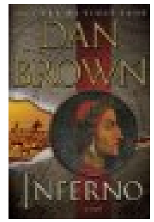


The Fast Metabolism Diet: E...
Hardcover
~~$26.00~~ $14.77



Inferno: A Novel (Robert La...
Hardcover
~~$29.95~~ $17.49

Best Sellers of 2013 (So Far) › See Top 100

StrengthsFinder 2.0
Hardcover
~~$24.95~~ $13.39

Lean In: Women, Work, and t...
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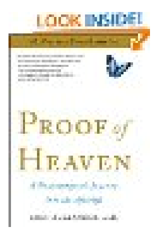
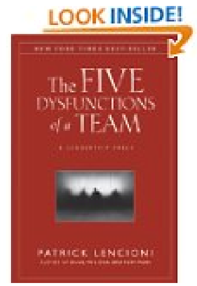
~~$24.95~~ $13.72



Proof of Heaven: A Neurosur...
Paperback
~~$15.99~~ $8.79



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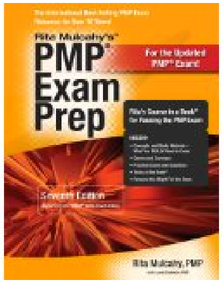

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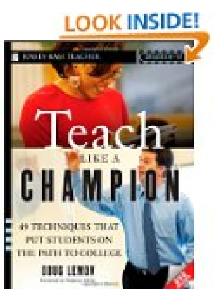

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Implications for Pricing?

- Gains
  - Segregate gains because the gain function has diminishing returns
  - Segregate small gains (silver linings) from larger losses
- Losses
  - Integrate losses because the loss function has diminishing loss
  - Integrate smaller losses with larger gains to offset loss aversion

Source: Thaler, R. (1999). “Mental Accounting Matters,” *Journal of Behavioral Decision Making*, 12, 183-206.


IT'S HERE.
Our new collection includes
14 new and exciting styles
Victor Jeffreys, filmmaker
Winston, Revolver Black

Price as a Signal of Quality

The four founders of Warby Parker (including me) were students at Wharton at the time we launched the company. Seeking advice about pricing, we approached our marketing professor Jagmohan Raju with a PowerPoint presentation that outlined our strategy. The plan, we proudly explained, was to sell prescription glasses that typically cost $500 for $45 — and to do so online. Without even glancing at our slides, Professor Raju shook his head. “No way,” he said. “It won’t work.”

No customer would trust that our quality was even comparable, much less equal or better.

We surveyed friends and acquaintances and found that a willingness to purchase glasses online increased with increases in price up to $100, at which point it plateaued and ultimately decreased. Price was indeed an indicator of quality. For many customers, $100 was the critical threshold.

Price as a Signal of Quality

- Under what circumstances
  - Consumers haven't experienced the product
  - Have a reference value to compare to
    - Consumers don't arbitrarily compute value from price. Rather, they compare the price to a product in the category for which they know the quality
- Experience and credence goods

Price as a Signal of Quality

- What if consumers actually get more utility from a purchase when its price goes up
  - "Quality" is never completely resolved
  - Conspicuous consumption
- It's no longer true that experience alone will dictate whether this effect holds
- This is a world where up is down and down is up

Price as a Signal of Quality

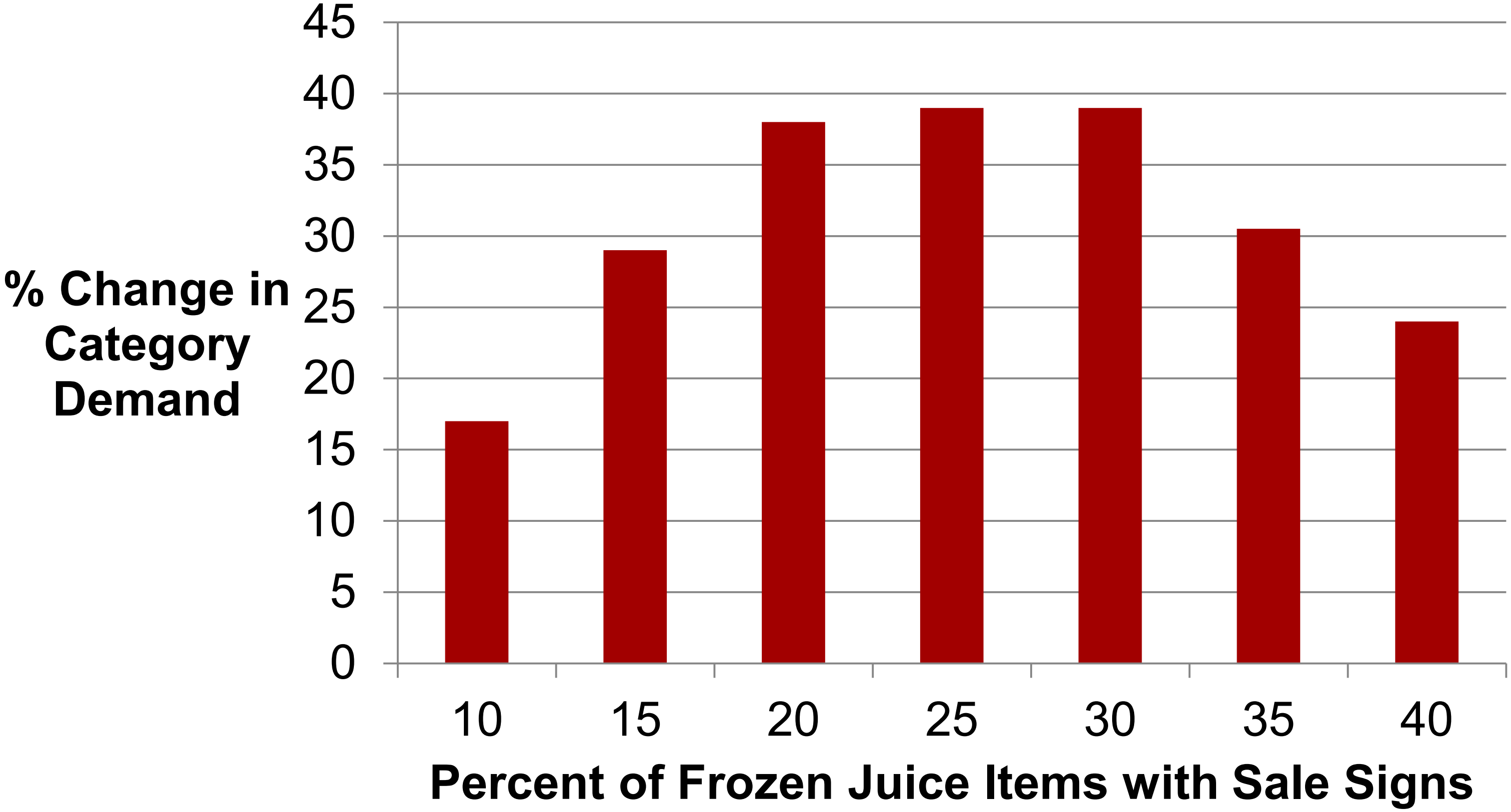
Which wine bottle has the largest markup?

The Second-Cheapest Bottle Of Wine Has The Highest Mark-Up & Other Restaurant 'Secrets'

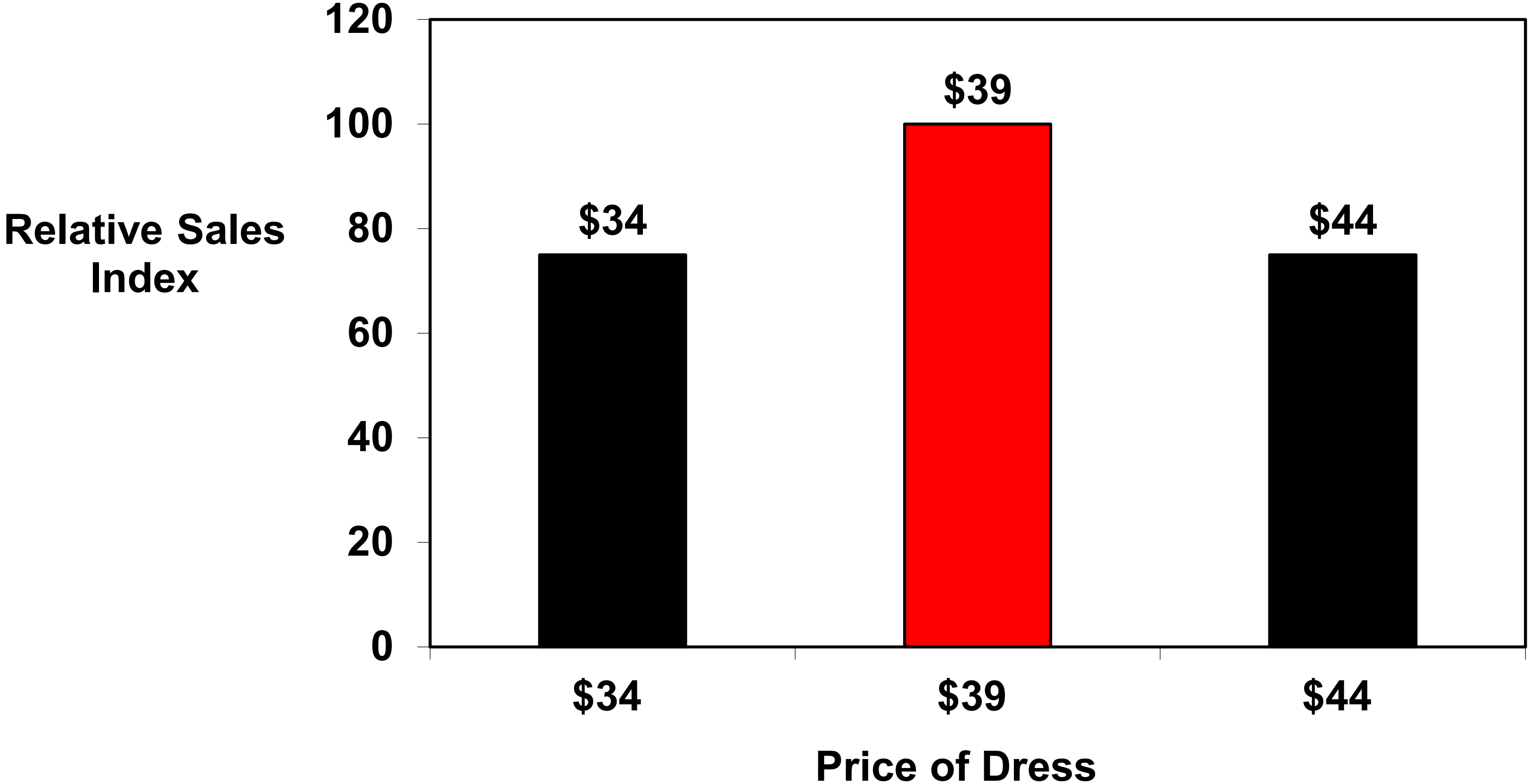
By Chris Morran December 6, 2013

Pricing Cues: Sale Signs

- Consumers infer whether or not a price is a good deal based on firm-provided cues:



Pricing Cues: Effect of a "9"



Pricing Cues: Effect of a “9”

- Two possible explanations:
  - Consumers “code” the 9 with the lower $10 increment
  - Stores use prices that end in 9 for discounts and consumers respond to that
- Conversely:
  - Signaling quality with a product that ends in 0

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SHOP ALL

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$1,695.00

SEE VIEWS ›

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SEE VIEWS ›

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SEE VIEWS ›

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$1,095.00

DRESS SHIRTS

SHOP ALL

SEE VIEWS ›

BOSS Black Sharp Fit Dress Shirt
$175.00

SEE VIEWS ›

Thomas Pink Slim Fit Dress Shirt
$185.00

SEE VIEWS ›

David Donahue Trim Fit Dress Shirt
$135.00

SEE COLORS & VIEWS ›

David Donahue Regular Fit Dress Shirt
$135.00

Behavioral Pricing: Summary

- Think about gains and losses in terms of prospect theory
  - Framing, loss aversion, and reference prices
- Price as a signal of quality
  - Especially relevant when consumers haven't yet experienced the product (of course has limits)
- Price cues (sale signs, prices ending in '9') can be important
  - But can also be overused

Pricing Takeaways

- Optimal pricing
  - Price elasticity (WTP), Competition, Marginal Cost
  - Value Based Pricing
    - Can use "engineering" estimate or WTP for attributes from conjoint
  - Rules of thumb
    - Cost + ➔ No!
    - Remember cost savings end up costing you money!
  - Price elasticity (WTP), Competition, Marginal Cost
  - Behavioral Pricing
    - Think in terms of Prospect Theory

Pricing Takeaways

- Walked through the mechanics of choosing an optimal price
  - This is what you will actually do!
- Optimal pricing necessarily excludes some people from the market
  - Otherwise this would be easy (set p=mc)
  - The key is to figure out how many and which ones
- Set prices by best responding to how you think your competitors will act
- Carefully integrate behavioral pricing techniques
  - Sometimes it's not clear how to balance them off against optimal pricing considerations
  - Experiments can be helpful here

Assignments for Tomorrow

- Case memo & discussion: Food Delivery
  - Why are restaurants unhappy?
  - Should we adjust pricing strategy to fix it? If so, how?

- Assignment: Optimal Pricing Using Data (Individual)